

# Model Development Document

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Complaint → Regulation Classifier (CMPL-REG-24)  
First line of defense — Model Development

Generated: 2026-07-23 21:44 UTC  
LLM: nim (meta/llama-3.1-8b-instruct)  
Data: CFPB Consumer Complaint Database (real, redacted narratives)  
Dataset: 4000 rows · regulatory rate 0.875  
Repo: reg-agents · model id: CMPL-REG-24

# 1 · Executive summary & methodology

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## Executive Summary

This Model Development Document outlines the design, estimation, and validation of a two-stage complaint-to-regulation classification model. The model is intended to classify consumer complaints into one of several regulatory categories, with the ultimate goal of informing regulatory action. The model's performance is critical to ensuring that regulatory resources are allocated effectively and that consumers receive timely and appropriate assistance.

The model's two-stage architecture is designed to balance the trade-off between recall and precision. In the first stage, a cheap and efficient classifier (logistic regression) is used to screen out non-regulatory complaints, thereby reducing the computational burden and improving the model's overall recall. The second stage employs a more expensive and accurate retrieval-augmented generation pipeline (rag\_llm) to label the remaining complaints with high precision. This design choice is motivated by economic and statistical considerations, as the cost of misclassifying a regulatory complaint is significantly higher than the cost of misclassifying a non-regulatory complaint.

## Methodology Narrative

The model development process began with the selection of a suitable dataset, which consisted of 4000 consumer complaints from the CFPB Consumer Complaint Database. The dataset was split into training, validation, and test sets, with 3040, 380, and 380 complaints, respectively. The regulatory rate, which represents the proportion of complaints that are regulatory in nature, was found to be 0.875.

In the first stage, two classifiers were compared: logistic regression and xgboost. The logistic regression model, with a threshold of 0.397, achieved a validation PR-AUC of 0.9948 and a test PR-AUC of 0.992. The xgboost model, with a threshold of 0.575, achieved a validation PR-AUC of 0.9903 and a test PR-AUC of 0.9896. The logistic regression model was ultimately selected as the champion model, based on its superior performance on the validation set.

The confusion matrix for the logistic regression model is provided in the metrics section, with 41 false positives and 7 false negatives. The model's accuracy, precision, recall, and F1 score are also reported, with values of 0.9079, 0.9775, 0.9157, and 0.9456, respectively.

In the second stage, the rag\_llm pipeline was used to label the complaints that were not screened out by the first-stage classifier. The pipeline achieved an accuracy of 0.28 and a family accuracy of 0.41. The macro F1 score and weighted F1 score were also reported, with values of 0.2677 and 0.2345, respectively. The per-label recall values are also provided, with values ranging from 0.0 to 1.0.

## Known Weaknesses

One known weakness of the model is the use of weak labels in the second stage. The rag\_llm pipeline relies on the output of the first-stage classifier to generate labels, which may not be accurate. This can lead to biased estimates of the model's performance and may require additional validation and calibration.

## Monitoring and Guardrail Design

To ensure the model's performance and prevent potential biases, a monitoring and guardrail design is implemented. The model's performance is tracked on a regular basis, and any significant changes or

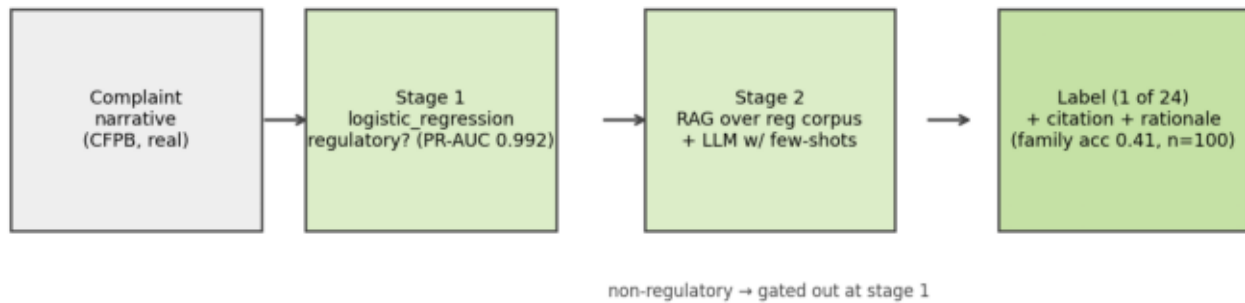
## **1 · Executive summary & methodology (cont.)**

deviations are investigated and addressed. Additionally, the model's output is subject to regular review and validation by subject matter experts to ensure that it is accurate and unbiased.

In conclusion, the two-stage complaint-to-regulation classification model is designed to balance the trade-off between recall and precision. The model's performance is critical to ensuring that regulatory resources are allocated effectively and that consumers receive timely and appropriate assistance. While there are known weaknesses, such as the use of weak labels, the model's design and implementation are intended to mitigate these risks and ensure that the model is accurate and unbiased.

## 2 · Architecture

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Two-stage pipeline: binary gate, then RAG+LLM labeling with citations.

## 2 · Architecture (detail)

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Stage 1 (binary gate): TF-IDF (1-2 grams, 30k features) into a logistic-regression vs XGBoost bake-off; champion selected on PR-AUC. Upgrade path: fine-tuned BERT-class encoder (NeMo Framework) served via Triton, same interface.

Stage 2 (multi-class): retrieval-augmented generation over the regulation/policy corpus (NeMo Retriever embeddings, FAISS locally / cuVS-Milvus on GPU) + LLM reasoning (NIM Llama-3.1-8B) with 8 curated few-shot examples and the 24-category taxonomy in-prompt. Output is strict JSON: label, confidence, rationale, and the cited source document, which the UI renders alongside the excerpt. A keyword scorer provides a deterministic no-LLM fallback; stage-1 gates non-regulatory complaints so the LLM is only invoked when needed.

Deployment: complaint MCP server (tools: `classify_complaint`, `sample_complaints`, `get_model_metrics`) + Complaint A2A agent; wired into the Streamlit UI and observable via the existing Prometheus/Grafana and OpenTelemetry stack.

## 3 · Data & curation

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Source: real, redacted consumer complaint narratives from the public CFPB Consumer Complaint Database (4000 curated rows; 200 reserved as a 5% scoring holdout for the batch-ingestion layer before any modeling split, then 3040 train / 380 validation / 380 test; regulatory rate 0.875).

Curation (scripts/fetch\_cfpb\_complaints.py) mirrors NVIDIA NeMo Data Curator stages: length filtering (ScoreFilter), exact deduplication (ExactDuplicates), near-deduplication (FuzzyDuplicates/MinHash analog), PII verification (CFPB pre-masks PII as XXXX; PiiModifier analog), and per-issue balanced sampling. At corpus scale each stage maps directly to the GPU-accelerated Curator module.

Ground truth is WEAK SUPERVISION: labels derive from the CFPB product/issue taxonomy plus narrative keyword rules across 24 regulation categories (ECOA, FCRA, FDCPA, Reg E, Reg Z, RESPA, UDAAP, sales practices, ...). This is the standard bootstrap before human-adjudicated labels exist and is flagged as a limitation in the validation report.

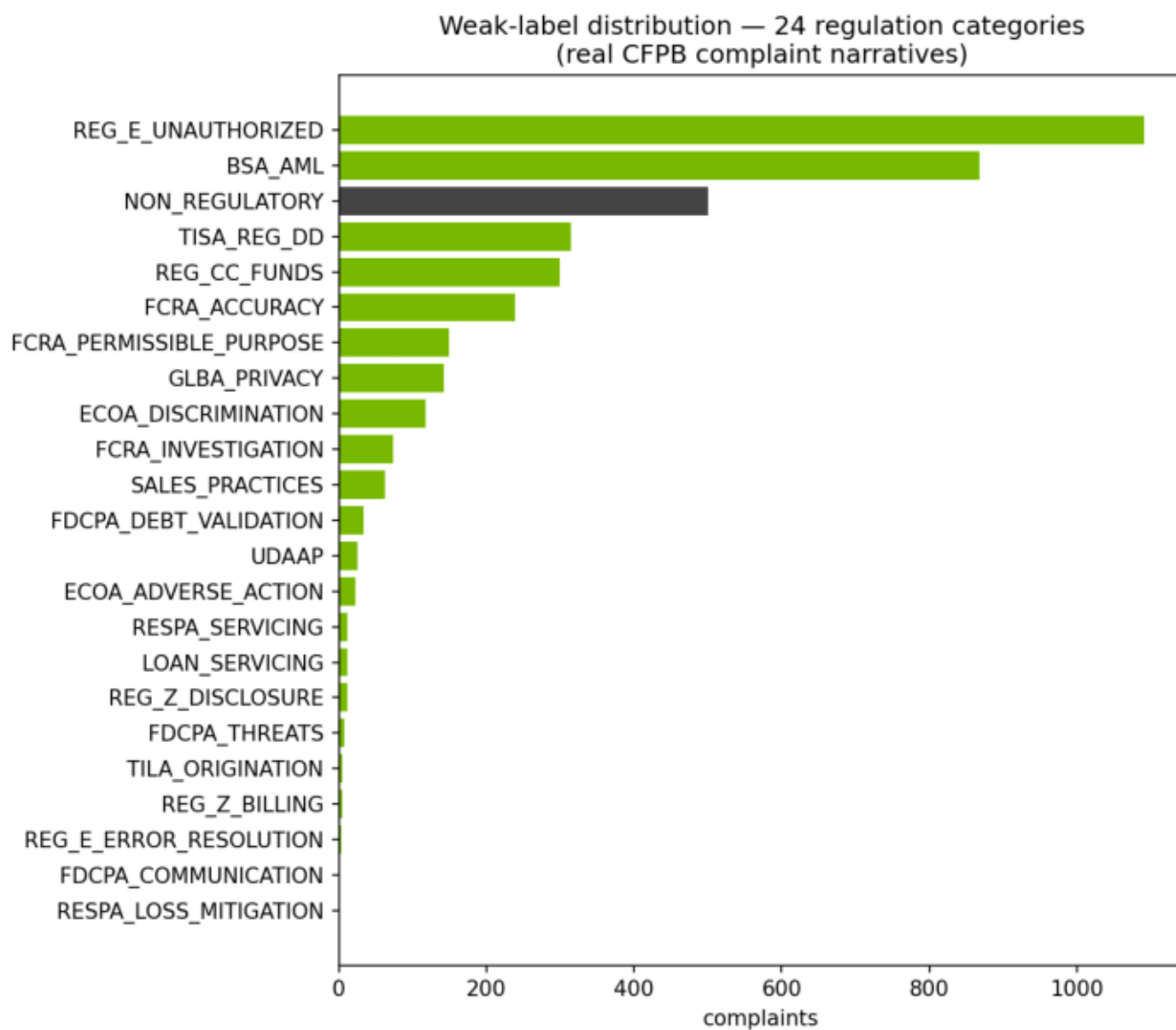
### 3 · Dataset summary

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| property                         | value                                                                    |
|----------------------------------|--------------------------------------------------------------------------|
| source                           | CFPB Consumer Complaint Database (public, PII-redacted)                  |
| curated rows                     | 4,000                                                                    |
| scoring holdout (reserved first) | 200 (5%, stratified — fed only through the ingestion layer)              |
| train / val / test split         | 3,040 / 380 / 380 (stratified 80/10/10 of the remaining 95%)             |
| regulatory rate                  | 87.5%                                                                    |
| taxonomy coverage                | 23 of 24 categories                                                      |
| curation stages                  | length filter · exact dedup · near dedup · PII check · balanced sampling |
| label provenance                 | weak supervision (CFPB issue taxonomy + keyword rules)                   |

### 3 • Label distribution

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Weak-label distribution over the 24 regulation categories.



### 3.1 · Regulation taxonomy (with dataset support)

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| code                     | regulation / category                                   | n in dataset |
|--------------------------|---------------------------------------------------------|--------------|
| FCRA_ACCURACY            | FCRA — Accuracy of Reported Information                 | 239          |
| FCRA_INVESTIGATION       | FCRA — Reinvestigation of Disputes                      | 74           |
| FCRA_PERMISSIBLE_PURPOSE | FCRA — Permissible Purpose / Improper Use               | 149          |
| FDCPA_DEBT_VALIDATION    | FDCPA — Debt Validation / Not Owed                      | 34           |
| FDCPA_COMMUNICATION      | FDCPA — Communication Tactics                           | 1            |
| FDCPA_THREATS            | FDCPA — False Statements or Threats                     | 7            |
| REG_E_UNAUTHORIZED       | Reg E / EFTA — Unauthorized Transfers                   | 1091         |
| REG_E_ERROR_RESOLUTION   | Reg E / EFTA — Error Resolution                         | 3            |
| REG_Z_BILLING            | Reg Z / FCBA — Billing Error Disputes                   | 5            |
| REG_Z_DISCLOSURE         | Reg Z / TILA — Fees, Interest & Disclosures             | 11           |
| TILA_ORIGINATION         | TILA — Credit Origination & Underwriting Disclosures    | 5            |
| ECOA_DISCRIMINATION      | ECOA / Reg B — Credit Discrimination                    | 118          |
| ECOA_ADVERSE_ACTION      | ECOA / Reg B — Adverse Action Notices                   | 23           |
| RESPA_SERVICING          | RESPA — Mortgage Servicing & Escrow                     | 12           |
| RESPA_LOSS_MITIGATION    | RESPA — Loss Mitigation & Foreclosure                   | 1            |
| TISA_REG_DD              | TISA / Reg DD — Deposit Account Disclosures             | 315          |
| REG_CC_FUNDS             | Reg CC — Funds Availability                             | 300          |
| BSA_AML                  | BSA/AML — Account Freezes & Closures                    | 868          |
| GLBA_PRIVACY             | GLBA / Privacy — Information Sharing & Safeguards       | 143          |
| UDAAP                    | UDAAP — Unfair, Deceptive, or Abusive Acts or Practices | 26           |
| SALES_PRACTICES          | Sales Practices — Unauthorized Accounts/Products        | 63           |
| SCRA_MLA                 | SCRA / MLA — Servicemember Protection                   | 0            |
| LOAN_SERVICING           | Loan Servicing (Auto/Student/Personal) — UDAAP & S      | 12           |
| NON_REGULATORY           | Non-Regulatory — General Service                        | 500          |

## 4 • Stage-1 bake-off leaderboard

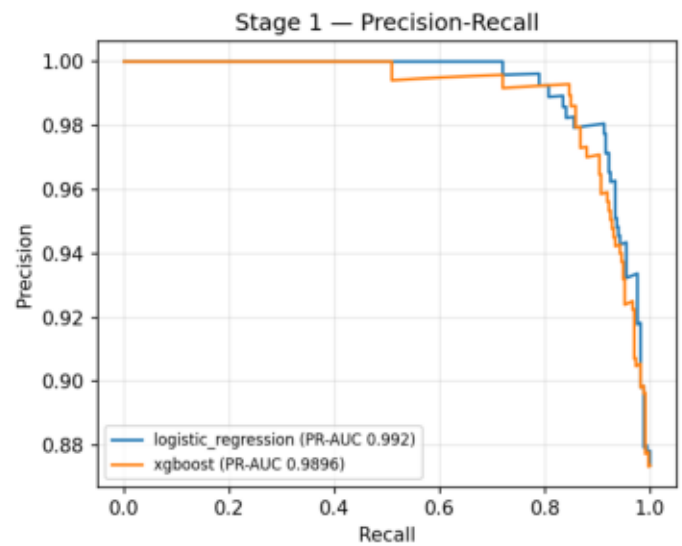
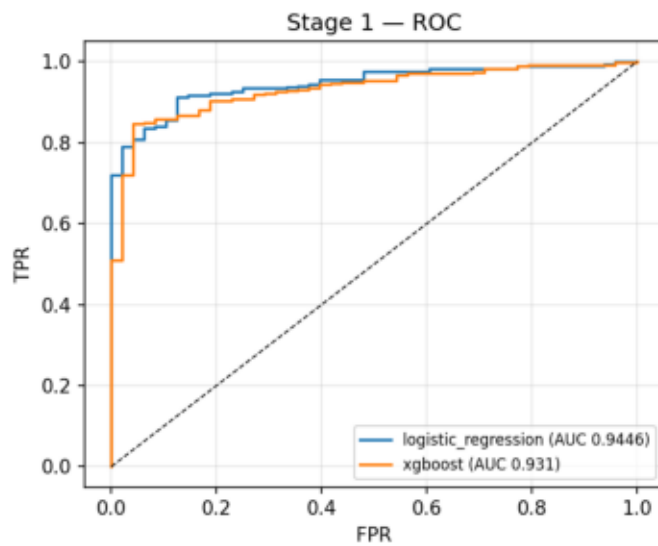
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|                | model               | params                 | val_pr_auc | threshold | pr_auc | train_roc_auc | roc_auc | train_test_gap | f1     | precision | recall | accuracy |
|----------------|---------------------|------------------------|------------|-----------|--------|---------------|---------|----------------|--------|-----------|--------|----------|
|                | logistic_regression | l1, C=2.0              | 0.9948     | 0.397     | 0.992  | 0.9751        | 0.9446  | 0.0305         | 0.9456 | 0.9775    | 0.9157 | 0.9079   |
| n_estimators=3 | logistic_regression | l1, C=2.0, max_depth=3 | 0.9903     | 0.575     | 0.9896 | 1.0           | 0.931   | 0.069          | 0.9335 | 0.9587    | 0.9096 | 0.8868   |

Champion: logistic\_regression - selected on validation PR-AUC; other columns are one-shot test metrics.

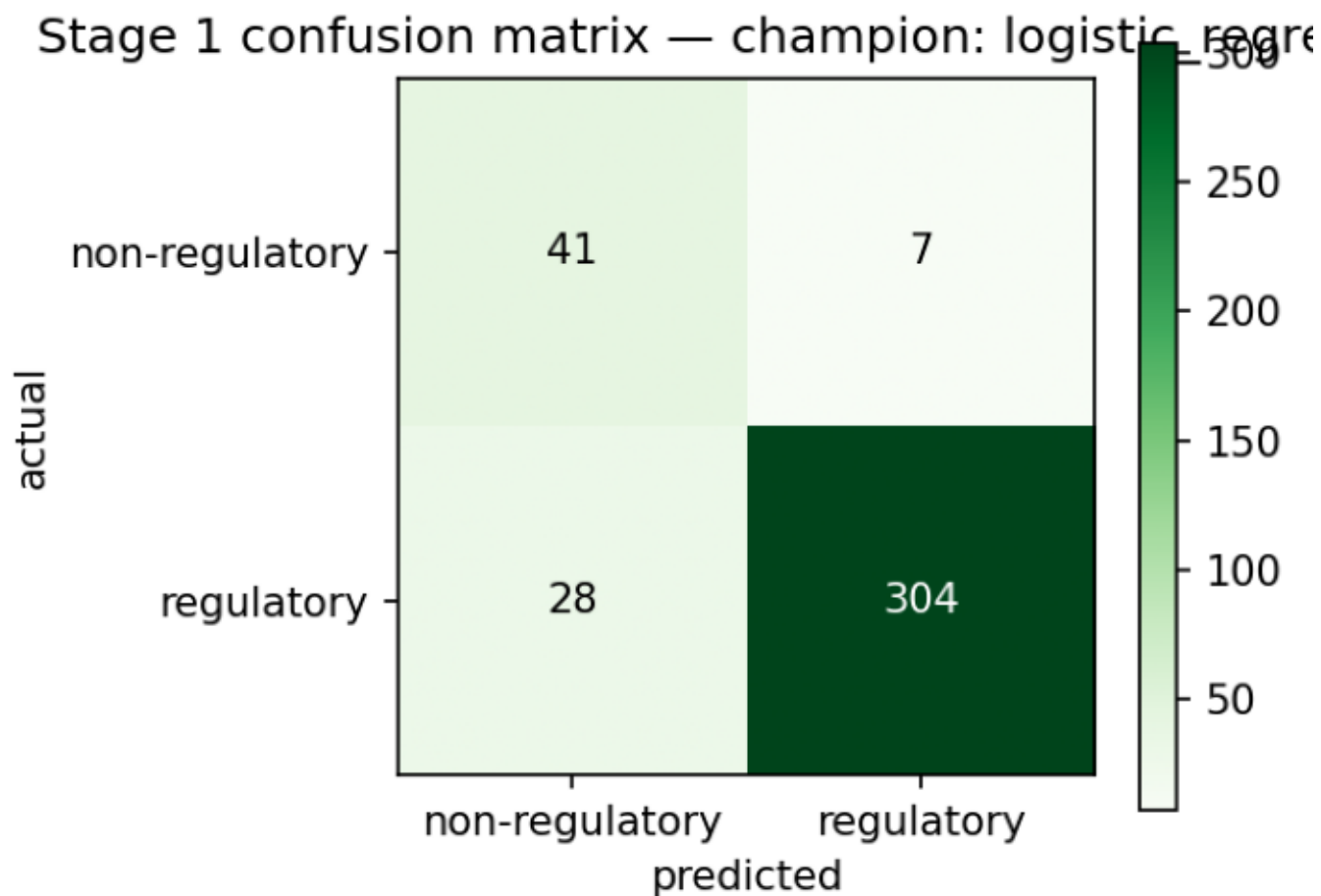
## 4 • Stage-1 ROC / PR curves

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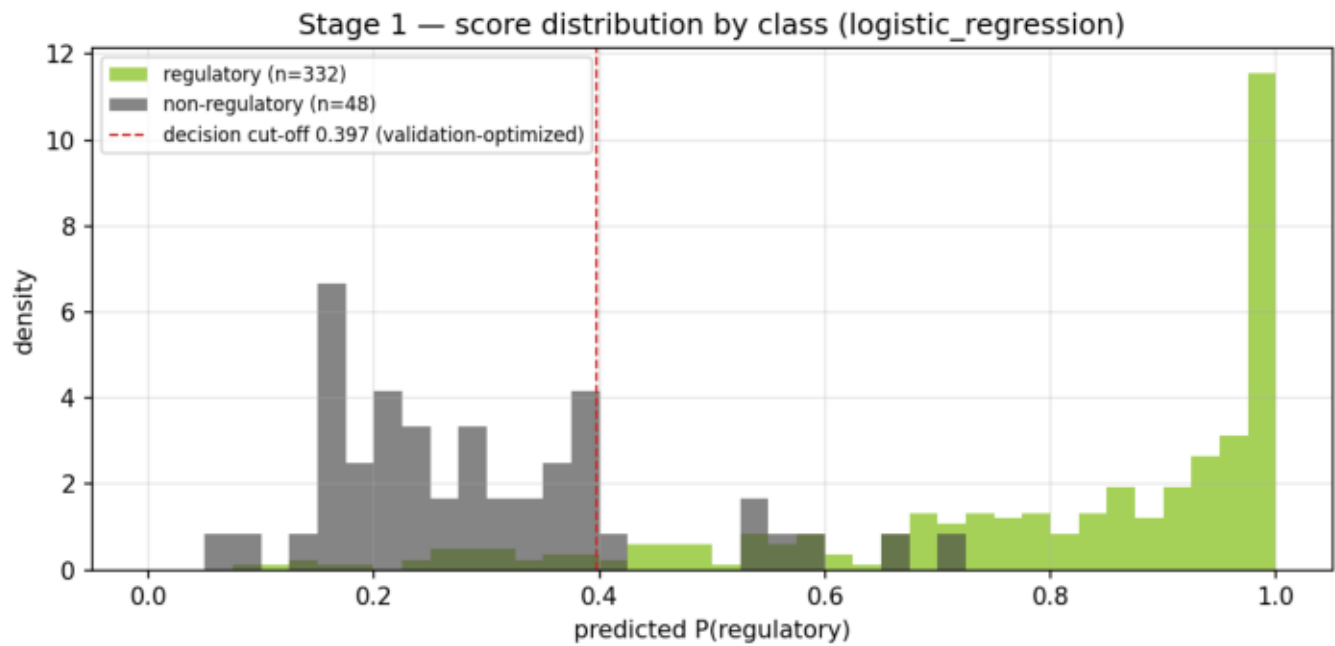
## 4 • Stage-1 confusion matrix

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## 4.1 · Score separation

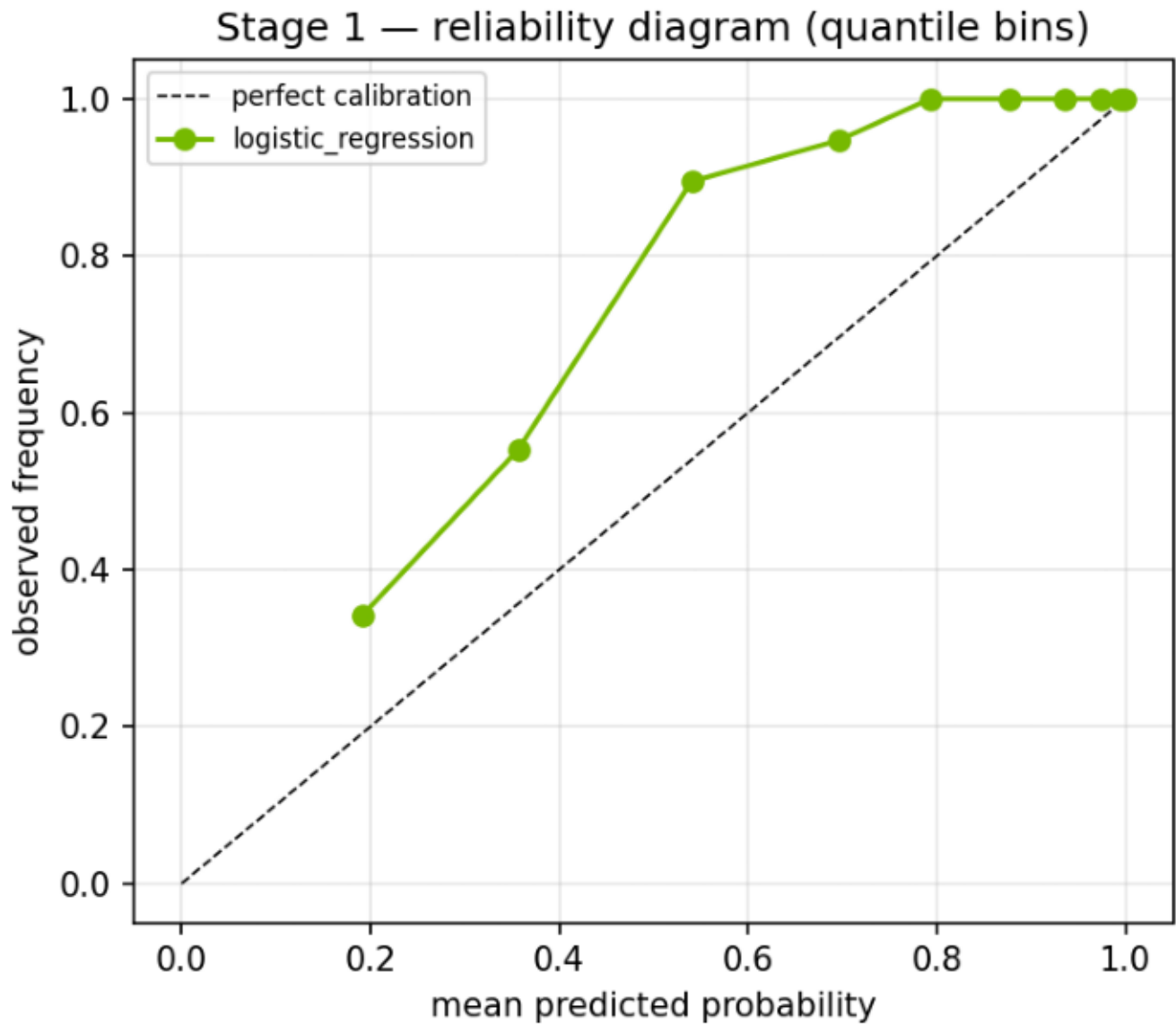
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Class separation at the validation-optimized cut-off 0.397 (default 0.5 not used).

## 4.1 · Calibration (reliability diagram)

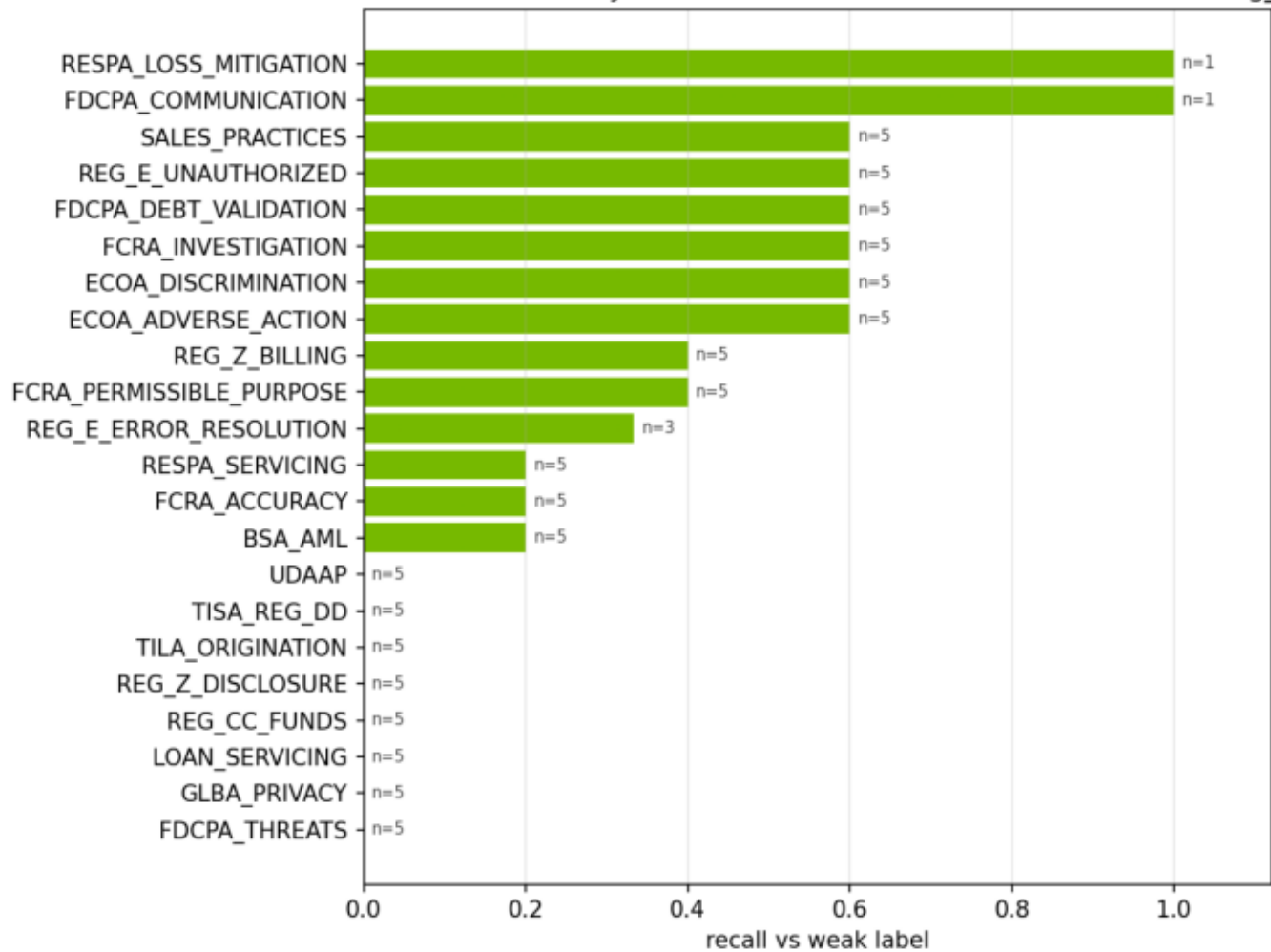
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Whether predicted probabilities can be read as probabilities.

## 5 · Stage-2 per-category recall

Stage 2 (RAG + LLM) — per-category recall  
exact acc 0.28 · family acc 0.41 · macro-F1 0.27 · n=100 · mode=rag\_ll



Exact acc 0.28 · family acc 0.41 · macro-F1 0.2677 · n=100.

## 5 • Stage-2 per-category detail

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| category                 | support | recall |
|--------------------------|---------|--------|
| BSA_AML                  | 5       | 0.2    |
| ECOA_ADVERSE_ACTION      | 5       | 0.6    |
| ECOA_DISCRIMINATION      | 5       | 0.6    |
| FCRA_ACCURACY            | 5       | 0.2    |
| FCRA_INVESTIGATION       | 5       | 0.6    |
| FCRA_PERMISSIBLE_PURPOSE | 5       | 0.4    |
| FDCPA_DEBT_VALIDATION    | 5       | 0.6    |
| FDCPA_THREATS            | 5       | 0.0    |
| GLBA_PRIVACY             | 5       | 0.0    |
| LOAN_SERVICING           | 5       | 0.0    |
| REG_CC_FUNDS             | 5       | 0.0    |
| REG_E_UNAUTHORIZED       | 5       | 0.6    |
| REG_Z_BILLING            | 5       | 0.4    |
| REG_Z_DISCLOSURE         | 5       | 0.0    |
| RESPA_SERVICING          | 5       | 0.2    |
| SALES_PRACTICES          | 5       | 0.6    |
| TILA_ORIGINATION         | 5       | 0.0    |
| TISA_REG_DD              | 5       | 0.0    |
| UDAAP                    | 5       | 0.0    |
| REG_E_ERROR_RESOLUTION   | 3       | 0.333  |
| FDCPA_COMMUNICATION      | 1       | 1.0    |
| RESPA_LOSS_MITIGATION    | 1       | 1.0    |

Recall vs weak labels on the stratified evaluation sample.



## 6 · Guardrail inventory

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| guardrail            | mechanism                                      | surfaced as                             |
|----------------------|------------------------------------------------|-----------------------------------------|
| Taxonomy whitelist   | LLM label must be one of the 24 codes; else re | complaint_classifications_total{label}  |
| Strict-JSON parsing  | malformed LLM output -> deterministic keyword  | mode=fallback counter (alertable spike) |
| Stage-1 gating       | non-regulatory volume never reaches the l      | cost + attack-surface control           |
| Retrieval grounding  | answer must cite a retrieved corpus exce       | citation attached to every prediction   |
| Provider portability | OpenA <-> NIM via config; keyword mode if b    | mode label per classification           |

## Sign-off

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Prepared and reviewed under the bank's model risk management framework (SR 11-7 / OCC 2011-12).

Author, development document: Senior Model Developer & Quantitative Lead — PhD (Econometrics), 20 years in model development across credit, capital planning, fraud, and NLP.

Author, validation report: Senior Independent Model Validator — PhD (Econometrics), 20 years in second-line validation; independent of the development team per SR 11-7 organizational-independence requirements.